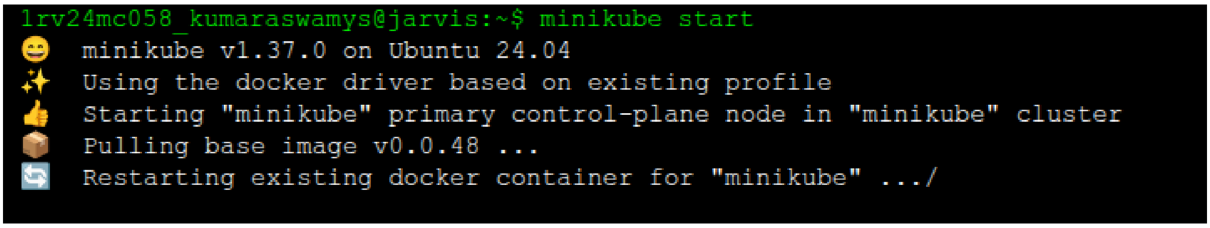
Program 8: Create Kubernetes Pods using YAML Descriptors

Project Structure

program8
|--nodejs-configmap.yaml
|--nodejs-pod.yaml
|--server.js

STEP 1 - Start Minikube

minikube start



```
1rv24mc058_kumaraswamys@jarvis:~$ minikube start
😄  minikube v1.37.0 on Ubuntu 24.04
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
📦  Pulling base image v0.0.48 ...
🔄  Restarting existing docker container for "minikube" .../
```

STEP 2 - Check Minikube Status

minikube status
You should see:
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

```
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```

Create a node js app

Nano server.js

```
const express = require('express')
const app = express()
const port = 3000

app.get('/', (req, res) => {
  res.send('Hello World!')
})

app.listen(port, () => {
  console.log(`Example app listening on port ${port}`)
})
```

STEP 3 - If Minikube is not running, restart it
minikube delete
minikube start --driver=docker

STEP 4 - Create the Deployment file (nodejs-deployment.yaml)
Create a new file:
nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
name: nodejs-deployment
spec:
replicas: 2
selector:
matchLabels:
app: nginxtemplate:
metadata:
labels:
app: nginxspec:
containers:
- name: node
image: node:latest
ports:
- containerPort: 3000

STEP 5 - Apply the deployment
kubectl apply -f node-deployment.yaml

STEP 6 - Check Deployment & Pods
kubectl get deployments
kubectl get pods
You should see 2 pods running.

STEP 7 -Expose the Deployment as a Service
kubectl expose deployment node-deployment \
--type=NodePort \
--port=80
This creates a service that exposes node to your system.

STEP 8 - Check Services
kubectl get services

```
1rv24mc058_kumaraswamys@jarvis:~$ kubectl get services
NAME              TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)          AGE
kubernetes        ClusterIP   10.96.0.1                    443/TCP          8m12s
node-deployment  NodePort    10.101.24.25                 80:30269/TCP     11s
```

A service named node-deployment should appear .

STEP 9 - Access the Webpage
Minikube provides a shortcut to open the app:minikube service node-deployment

STEP 10 — Clean Up
Delete all pods
kubectl delete pods --all
Delete all services
kubectl delete svc --all
Delete all deployments
kubectl delete deployments --all